NOCO Asset IQ: Master Briefing

Nov 2, 2026 · @Lavya

Start here

NOCO Asset IQ is a web app that tells a commercial building owner which energy upgrades to make, what order to make them in, and what that order is worth. It works from a few minutes of building data, before any contractor visit.

The one-line pitch: every upgrade, in the right order.

The 20-second version: most retrofits are planned one upgrade at a time, so owners buy HVAC sized for a building they are about to improve. Asset IQ tests every combination and order of five upgrades against the owner's budget, and shows the plan worth the most. On our sample Buffalo office, the same three upgrades in the worst order are worth $69,476 less and go $16,249 over budget.

If you built slides or notes before November 2, update them. The model was rebuilt on sourced costs, so the demo's numbers changed a lot:

Balanced path on the demo	Old	Now
Budget	$250,000	$600,000
Order	Envelope → LED → HVAC	LED → Envelope → HVAC
Net investment	$246,561	$579,335
Savings per year	$48,939	$47,627
Simple payback	5.0 years	12.2 years
Saved by sequencing	$18,004	$36,915
Data score	90% High confidence	90% data provided

The app also gained a fourth path (Maximum Savings), the Order Lab, and a 5.2-year "payback on the extra spend" figure.

How to use this document: read it once top to bottom tonight. Then rehearse with the demo script and the hard questions, which are the two sections you will use most.

- Live app: lavya-gohil.github.io/noco-pathfinder
- Straight to the sample results: the #/demo link

The problem and the idea

Upgrades change each other, so doing them in the wrong order wastes money, and adding up their savings overstates the result. Most planning ignores both.

Three things go wrong today:

- **Equipment sized for the old load.** Replace HVAC first and you pay for capacity the building stops needing once the envelope is fixed.
- **Savings that add up too well.** Upgrades overlap, so a simple sum counts the same kilowatt-hour twice. On the demo, a simple sum claims $50,014 a year and the real figure is $47,627.
- **Budgets spent in the wrong place.** A fixed budget buys less without sequencing. On the demo, the Balanced plan costs $579,335 in the right order and $616,249 big-equipment-first, so it only fits a $600,000 budget when sequenced.

Who it's for: the owner or manager of a commercial building (office, warehouse, retail, school or multifamily) who is deciding where to spend capital, before paying for an audit.

What it is not: an energy audit, an engineering design or a quote. It's a pre-audit screen for comparing options and deciding what to verify on site. Say this before someone else does. It makes everything else more credible.

NOCO and the business case

The pitch to NOCO is that this could help a sales team turn one building's data into a staged series of jobs, instead of a single quote.

What we found about NOCO, mostly from its own marketing and search results, so treat it as a starting point:

- A family-owned energy company based in Tonawanda, NY, founded in 1933 as a coal supplier, serving Western and Central New York.
- Its commercial pages describe HVAC installation and service (furnaces, boilers, heat pumps, rooftop units), plus fuels and natural gas and electricity supply. Solar and generators also appear in its listings.

Why the demo fits: the sample is a Buffalo office with old rooftop units and a gas furnace, which is squarely the kind of customer those pages describe.

The argument to make:

1. A roadmap gives the customer a reason to start with a small first phase, and a reason to come back for the next ones.
2. It reaches the owner before a competitor's single-equipment quote does.
3. NOCO's real project costs and results could replace our planning figures. That would make the tool much more accurate, and it's something only NOCO could do.

Confirm before you say it:

- [ ] Check NOCO's actual commercial service list. We couldn't confirm it fully, so nothing in the app claims which phases NOCO can deliver. In particular, we did not confirm LED retrofits, controls or envelope work.
- [ ] Decide how to describe the relationship. The app carries NOCO's name on a public site, and now says "A concept by Axion Dynamics, prepared for NOCO." Keep that line unless NOCO has already agreed to its name being used.

We found nothing about any formal NOCO competition or its judging criteria, so if there is a brief or rubric, follow that over this document.

Product tour

The app runs in four steps, Building → Diagnosis → Strategy → Report, and the Strategy step holds the most important screens.

Screen	What it shows	Why it matters
Homepage	The pitch, a live sample result, how it works	Figures come from the same engine as the app
Building (4 sections)	Property, systems, energy bills, budget and objective	Takes about two minutes. Year built, HVAC year, kWh and monthly usage are optional
Diagnosis: Energy profile	Spend, cost per sq ft, breakdown by end use	The baseline every upgrade is measured against

Screen	What it shows	Why it matters
Diagnosis: Data completeness	How much useful data was given, and the next most valuable data point	Measures the inputs, not the accuracy of the estimates
Diagnosis: Signals	Patterns worth checking on site, like aging HVAC	Points the site visit
Strategy: Choose a path	Four plans side by side, with a live budget slider	Every path re-optimizes as the budget moves
Strategy: Roadmap	Phases in order, plus the strategy summary and the "HVAC is due anyway" comparison	Holds the 5.2-year figure
Strategy: Why this order	The reasoning, plus the **Order Lab**	The live demo moment
Strategy: Scenarios	Budget, electricity price and incentive sliders, plus a cash flow chart	Tests the plan against what could change
Strategy: Compare measures	Each upgrade alone, with its assumptions in a side panel	Answers "where did that come from?"
Report	A printable executive roadmap	What an owner would take away

The four paths:

Path	Picks the plan with	Objective it matches
Quick Wins	The fastest payback, favoring bigger savings among near-ties	Fastest Payback
Balanced	The best mix of savings, payback and sequencing, keeping each phase within a 10-year payback	Balanced
Maximum Savings	The most dollars saved per year	Maximum Savings

Path	Picks the plan with	Objective it matches
Deep Retrofit	The largest energy and emissions reduction	Deep Retrofit

HVAC past its 20-year service life is exempt from Balanced's 10-year limit, because it needs replacing regardless.

Other features worth knowing:

- **Guided tour:** 21 steps in four chapters, where the user does the clicking and an animated arrow points at each control to use. The ? button opens the chapter for the current page.
- **Assumptions panel:** the button at the top of the Strategy page shows every input and estimate, and links to the sourced list.
- **Saved in the tab:** a reload keeps the assessment. Closing the tab deletes it.
- **Light and dark themes:** set with the icons at the top right.

The demo building's numbers

Know these cold: the Balanced plan costs $579,335 and saves $47,627 a year, and the number to lead with is the 5.2-year payback on the spend beyond replacing the HVAC. All figures are illustrative pre-audit estimates with a 15% incentive assumption.

The building: Riverside Commerce Center, a fictional 85,000 sq ft office in Buffalo, NY, built in 1988. It has a gas furnace and rooftop units with HVAC installed in 1994, so about 32 years old. It spends $172,000 a year on energy ($118,000 electricity on 720,000 kWh, and $54,000 heating), and has a $600,000 budget.

The recommended plan, Balanced:

Phase	Upgrade	Net cost	Savings per year
1	LED lighting	$108,375	$17,911
2	Envelope and insulation	$77,067	$13,304
3	High-efficiency HVAC	$393,893	$16,412
	Total	**$579,335**	**$47,627**

That's $681,570 gross less $102,236 of potential incentives, a 28% cut in energy spend and 148 tonnes of CO_2e a year.

The three numbers that tell the story:

Number	What it means	Where to show it
$36,915	HVAC net cost falls from $430,807 to $393,893 when LED and envelope come first	Why this order
5.2 years	Beyond replacing the HVAC alone ($430,807, saving $19,113 a year), the plan adds $148,527 for $28,514 more a year	Roadmap
$69,476	The value gap between the best order and the worst. The worst also goes $16,249 over budget	Order Lab

Don't mix these up:

- **12.2 years** is the plain payback on the whole plan, and **5.2 years** is only on the extra spend. Both are true, so always say which one you mean.
- **−$111,720** is the plan's 15-year value at a 6% discount rate. It's negative because it charges $350,563 in today's dollars for an HVAC replacement that's due anyway.
- **$62,070** compares our order with big-equipment-first. **$69,476** compares it with the worst order. Pick one to quote.

All four paths at $600,000:

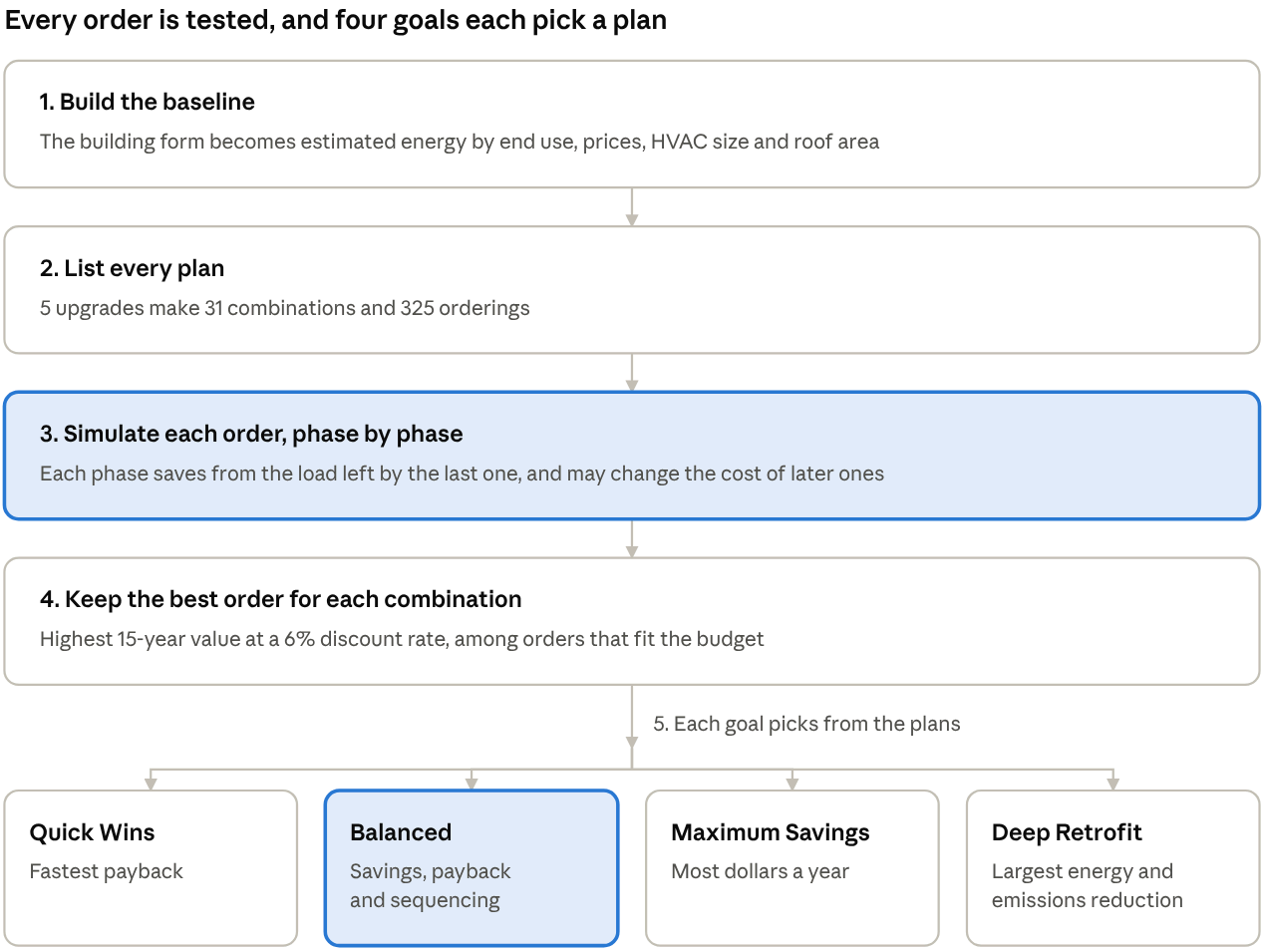
Path	Order	Net cost	Savings per year	Payback	HVAC
Quick Wins	LED → Envelope	$185,442	$31,215	5.9 years	Left in place
Balanced (recommended)	LED → Envelope → HVAC	$579,335	$47,627	12.2 years	Replaced
Maximum Savings	LED → Envelope → Solar	$543,608	$54,871	9.9 years	Left in place
Deep Retrofit	Envelope → Controls → Solar	$574,429	$51,563	11.1 years	Left in place

Each upgrade on its own:

Upgrade	Net cost	Savings per year	Payback
Envelope and insulation	$77,067	$12,990	5.9 years
LED lighting	$108,375	$17,911	6.1 years
Smart controls	$90,313	$13,247	6.8 years
Rooftop solar	$437,713	$28,910	15.1 years
High-efficiency HVAC	$430,807	$19,113	22.5 years

How the engine works

The engine simulates every order of every combination of five upgrades, keeps the most valuable order for each, then picks the best plan for each of four goals. There's no AI model in it: it's deterministic arithmetic, so the same inputs always give the same answer.



How Asset IQ turns building data into four plans

Step 3 is where the product's value lives: it's why the order can come out differently for different buildings. The objective the owner chose then decides which path is marked Recommended.

The five upgrades: envelope and insulation, LED lighting, high-efficiency HVAC, smart controls, and rooftop solar. That makes 31 combinations and 325 orderings.

How one simulation works: each phase saves energy only from the load left by earlier phases, never from the original baseline. That's what prevents double counting.

The interactions it models:

Interaction	Effect
Envelope before HVAC	Design load falls 3–15% depending on age and climate, so HVAC can be smaller. Only 70% of HVAC cost scales with size

Interaction	Effect
LED before HVAC	Less heat from lights trims cooling load and capacity by about 2%. It also adds some winter heating, which is taken off the savings
Controls with HVAC	Their overlap is discounted by 20%. If controls go first, 25% of their cost is spent again to recommission them
Efficiency with solar	The array is sized to the demand left after the other upgrades. It costs less and generates less, so this doesn't count as a saving

How it chooses an order: phases start a year apart, so delaying a fast-payback upgrade has a cost. Each order is valued over 15 years at a 6% discount rate, and the highest value that fits the budget wins. That's why the answer isn't a fixed rule: different buildings get different orders.

Two rules about equipment age: HVAC under 10 years old isn't recommended for replacement, unless it's electric resistance heat. HVAC 20 years or older is treated as due for replacement.

Assumptions and limits

The costs have real sources, but the savings percentages are engineering judgement that hasn't been calibrated against measured buildings. This is the model's biggest weakness, so own it openly. The full list with quality grades is in docs/assumptions.md.

Input	Value	Basis	Confidence
LED retrofit	$1.00–1.75 per sq ft	Between relamping and full redesign in a 2022 City of Seattle costing study	Moderate, with a scope judgement
Air sealing	$0.40 per sq ft of floor	Same study, $0.22–0.54	Moderate
Roof insulation	$2.00 per sq ft of roof	Same study, $0.19–3.86	Moderate, wide range

Input	Value	Basis	Confidence
Controls	$0.75–1.25 per sq ft	Same study, retro-commissioning at $0.50–2.00	Moderate
Packaged HVAC	$2,400 per ton	Contractor sites, mostly $2,000–4,000	**Weak**
Solar	$2.35 per watt	Berkeley Lab Tracking the Sun, 2017–2018 data	Moderate, dated
Value of a solar kWh	70% of average price	Solar cut demand charges by a median of 7–19%	Moderate
Savings percentages	Various	Engineering judgement	**Weak, uncalibrated**
Phase gap, discount rate, period	1 year, 6%, 15 years	Our choice, and they affect the order	**Judgement**
Incentives	15% of gross	Adjustable placeholder. No live programs are checked	**Placeholder**

Not in the model:

- Regional cost differences. The main cost source is one city in 2022.
- Real utility tariffs. Electricity is one average price, and demand charges only affect the solar value.
- Equipment failing before its planned phase. Failed equipment can't wait for a plan.
- Fuel switching from gas to heat pumps.
- Measure lifetimes beyond the 15-year window, which undervalues long-lived work like insulation.
- The lower true cost of efficiency in plan totals when equipment is due anyway. HVAC is carried at full replacement cost, which is conservative.

Tax rules changed recently. Based mostly on secondary summaries, the 179D deduction ended for construction starting after June 30, 2026. The federal solar credit now generally needs construction started by about July 2026 or service by the end of 2027. Don't quote these dates as definitive.

The presentation

The demo takes about four minutes and has one job: get them to see that order changes the money. This script assumes you start on the #/demo link. The slot is four minutes and there's now a PowerPoint, NOCO-Asset-IQ.pptx in the backup kit. Follow the timed script in its speaker notes rather than the table below, which runs slightly over and has no slides. The live Order Lab moment is the same in both.

Time	Do this	Say something like
0:00	Stay on the homepage or open with no screen	"A 32-year-old rooftop system in Buffalo needs replacing. Most owners just replace it. We think that's an expensive mistake, and we can show what it costs."
0:30	Open the #/demo link, on Choose a path	"We tested every combination and order of five upgrades against this owner's $600,000 budget. Here are the best plans for four different goals."
1:00	Drag the budget slider, then use the arrow keys to return to exactly $600,000. The mouse moves in jumps of about $15,000–20,000, and the numbers below depend on that exact budget	"Everything recalculates live, with no server and no data leaving the browser."
1:30	Continue to Roadmap, point at the summary	"12.2 years looks bad. But the HVAC was coming anyway. The real decision is the extra $148,527, and that pays back in 5.2 years."
2:30	Continue to Why this order, scroll to the Order Lab, click **Worst order**	Say nothing for a moment and let the numbers change. Then: "Same three upgrades, $69,476 worse, and now over budget."
3:15	Offer the controls to a judge: "Pick any order."	Let them try it. This proves the numbers aren't scripted.
3:45	Open the Report	"Before a contractor is called, the owner has a roadmap. For NOCO, that's a staged series of jobs instead of one quote."

Time	Do this	Say something like
4:15	Close	"This is a screen, not an audit. The savings factors need calibrating, and your project data is what would make it real."

How to split it between two people: one drives the screen and the other talks, so the speaker can watch the room. Agree who takes the technical questions and who takes the business ones.

If time is short, do only the Roadmap and the Order Lab. Those two moments carry the pitch.

If asked to try a real building, use Start my own assessment. The form is quick, and a live result on a building they name would be a strong moment. Rehearse it once first, and expect plain results if the data is thin.

Hard questions

NOCO's people know this field better than the model does, so the winning move is to name the weaknesses first and turn them into the reason to work together. Never defend a number you can't source. "That's a judgement call, and your data would replace it" is a strong answer.

They ask	Answer
Where do the costs come from?	A public 2022 Seattle costing study and Berkeley Lab data where possible, graded in our assumptions doc. HVAC per-ton pricing is the weakest, from contractor sites. NOCO's real numbers would replace it.
$2,400 a ton is wrong.	Probably, for some jobs. It's a single planning figure with no regional adjustment. What's your typical range? That's exactly the input we'd want from you.
Are the savings percentages validated?	No. They're engineering judgement and uncalibrated. Calibrating against NOCO's completed projects is the obvious next step.

They ask	Answer
A 12-year payback? Nobody buys that.	That's the whole plan, including an HVAC replacement that's due regardless. On the $148,527 beyond that, it's 5.2 years. If the owner isn't replacing HVAC, Quick Wins pays back in 5.9 years.
Why is the 15-year value negative?	It charges the plan $350,563 in today's dollars for HVAC that's coming anyway, and the app says so right beside it. Insulation also lasts well past the 15-year window, which we don't credit.
Why recommend Balanced when Maximum Savings looks better?	Maximum Savings costs less and saves more, but leaves a 32-year-old system in place, and that bill is still to come. The cards say so. An owner can choose either.
Why wait until year 3 to replace 32-year-old HVAC?	The model doesn't account for failure risk. If it's failing, it goes first, and the Order Lab shows what that costs: moving HVAC first adds $36,915. That's a useful number for that conversation.
How much does sizing really drop? Equipment comes in fixed sizes.	Good catch. The model treats cost as smooth, with 70% scaling with tons. Real savings would be lumpier, and a load calculation decides. It's a screening estimate.
Doesn't envelope work cost more than that?	It can. Published ranges are wide: roof insulation is $0.19–3.86 per sq ft in our main source, and we use $2.00. Scope varies a lot by building.
What about demand charges?	Not fully modeled. We use one average electricity price, and value solar kWh at 70% of it because solar cuts demand charges only modestly. Efficiency measures do reduce demand, but we don't model that separately.

They ask	Answer
What incentives does this assume?	A flat 15% placeholder, adjustable on the Scenarios page. It doesn't check live programs. Real NYSERDA or utility program data would be a clear upgrade. We haven't verified what's available today.
Is the electricity price right?	$0.164 per kWh comes from the sample building's own bills. It's illustrative. One source put New York's commercial average near 22 cents, though that isn't an authoritative figure.
Why no heat pump conversion from gas?	Fuel switching isn't modeled yet. Electric resistance to heat pump is. For an energy company, that's a natural next feature.
Why not use AI?	The result has to be explainable and repeatable. Every number traces back to a formula and an assumption you can inspect.
What happens to customer data?	It stays in the browser tab and is deleted when the tab closes. There's no server, database or analytics. A real product for NOCO would need accounts and storage, which is a design decision.
Is this your product or ours?	It's a concept prepared for you, and it isn't a NOCO product today. Make sure you can say who owns what, in line with the licence below.
What would you do next?	Calibrate against NOCO's project data, add real tariffs and incentive programs, model fuel switching and failure risk, then pilot with the sales team.

Tech and project facts

It's a browser-only React app with no backend, hosted on GitHub Pages, and backed by 108 automated tests.

Topic	Fact
Stack	React 19, TypeScript, Vite and Tailwind 4, with Recharts for charts
Repository	github.com/Lavya-Gohil/noco-pathfinder. It's public, so anyone can read the assumptions doc and the code
Engine code	`src/engine/` : `baseline.ts` , `measures.ts` , `interactions.ts` , `optimizer.ts` , `explanations.ts` , `confidence.ts`
Quality checks	108 tests and a CI workflow that runs lint, tests and build on every PR
Deployment	Merging doesn't update the site. Run the **Deploy to GitHub Pages** workflow in the Actions tab, or `npm run deploy`
Run locally	`npm install` , then `npm run dev`
Privacy	No server, accounts, cookies or analytics. The assessment lives in this tab's session storage and is deleted when the tab closes. The theme choice is saved in local storage
Licence	Proprietary, all rights reserved, held by Lavya Gohil. Open-source dependencies keep their own licences, listed in `third-party-licenses.md` in the built app
Assumptions	docs/assumptions.md, with every number's source and quality grade

A note on the licence: it's a template, not legal advice. Public code can still be viewed and forked on GitHub under GitHub's terms. Get a lawyer's review before relying on it commercially, or before discussing ownership terms with NOCO.

Tomorrow's checklist and backup plan

Rehearse on the actual presenting laptop tonight, with a timer. The app has only been tested in automated Chrome so far, never on a real projector.

Tonight:

- [ ] Update any slides or notes with the new numbers in Start here
- [ ] Rehearse the full demo twice, with a timer, on the presenting laptop
- [ ] Practice the hard questions on each other, out loud
- [ ] Check NOCO's service list and decide how you describe the relationship
- [ ] Agree who drives the screen, who speaks, and who takes which questions
- [ ] Copy the backup kit to the presenting laptop and test it

Just before you present:

- [ ] Open a fresh browser tab on the #/demo link. Don't reuse a tab with old work in it
- [ ] Set the **light theme** with the sun icon, since it should read better on a projector
- [ ] Set browser zoom so the Strategy page reads from the back of the room
- [ ] Close other tabs and silence notifications
- [ ] Have the backup ready and a phone hotspot available

If something fails:

Problem	Do this
No Wi-Fi, but the page was already loaded	Stay on the open tab. Pages already loaded keep working, but a page not yet visited may not load, so click through them all beforehand
No Wi-Fi at all	Serve the backup kit's `site` folder from the laptop, using the commands in its `README.txt`. Double-clicking `index.html` won't work
Nothing runs	Present from the kit's `report-sample.pdf` and `screenshots` folder
The app shows old data	Open a new tab. Each tab keeps its own saved state
You forget a number	Everything is on screen. Read it from the app rather than guess

The backup kit is at `C:\Users\vamsh\noco-demo-backup` on the machine this was built on, so it needs copying to the presenting laptop. Its Python command was tested on this machine on November 2, and the full demo ran from it. The npx command hasn't been

tested, and the presenting laptop needs Python or Node installed.

Sources

As of November 2, 2026. The model's full citations are in the assumptions doc.

Read directly:

- City of Seattle, Building Energy Efficiency and Electrification Costing Analysis, July 2022
- Berkeley Lab and NREL, Exploring Demand Charge Savings from Commercial Solar, executive summary
- NOCO, commercial electric page. It gave little detail on commercial services

Seen in search results only, so verify before quoting:

- NOCO Energy Corporation on Wikipedia, and NOCO's commercial heating and Buffalo pages
- Berkeley Lab, Tracking the Sun
- NREL, demand charges and storage
- PickHVAC and Oxmaint, for HVAC per-ton pricing
- IRS FAQs on 179D and BDO on the energy credit deadline
- Outage Pro, New York electricity rates